

INFORMATION NETWORK SECURITY ADMINISTRATION

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EMBEDDED SYSTEMS DEPARTMENT • SUMMER CAMP

SAFE LINK

Wearable-Based Group Safety and Participant Monitoring System

Project Proposal — Embedded Systems & IoT

Prepared for

INSA Embedded Systems Summer Camp Program

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Declaration

We, the members of the Embedded Systems Department summer camp project team, declare that this proposal, “SAFE LINK: Wearable-Based Group Safety and Participant Monitoring System,” is our own work, prepared under the guidance of our assigned camp mentors, and has not been submitted, in whole or in part, for any other project, award, or qualification.

All sources of information, technical specifications, and component references used in preparing this proposal have been acknowledged. Any figures, tables, or diagrams not otherwise credited are original work produced for this proposal.

_____

Team Representative — Name and Signature

_____

Date

Approval / Certification Page

This proposal, “SAFE LINK: Wearable-Based Group Safety and Participant Monitoring System,” submitted by the Embedded Systems Department summer camp project team, has been reviewed and is hereby submitted for evaluation and approval as a project proposal within the INSA Embedded Systems Summer Camp Program.

_____

Camp Mentor / Advisor — Name and Signature

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Date

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Department Head, Embedded Systems Department — Name and Signature

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Date

Acknowledgment

The project team gratefully acknowledges the Information Network Security Administration (INSA) and the Embedded Systems Department for the opportunity to develop this proposal as part of the summer camp program. We thank our mentors and camp coordinators for their technical guidance and feedback throughout the proposal-writing process.

We also acknowledge the open documentation provided by the manufacturers of the components referenced in this proposal, which informed the feasibility and component-selection sections of this document.

Abstract

SAFE LINK is a wearable-based safety and accountability system for organized outdoor groups — youth camps, university excursions, school field trips, hiking and scouting groups, outdoor training programs, and adventure tourism operators. Each participant wears a compact electronic device that reports status to a central system; organizers monitor the group through a digital dashboard rather than through continuous manual observation.

The system is built around a single design principle: surface exceptions, not raw data. Rather than asking an organizer to watch a stream of coordinates, heart rates, and battery levels, SAFE LINK converts sensor telemetry into a small set of operationally meaningful states — SAFE, WARNING, CRITICAL, and UNKNOWN — and only asks for attention when a participant's state changes for the worse. This allows a small organizing team to responsibly oversee a group far larger than they could track by headcount and radio alone.

The proposal covers eleven core features spanning identification, emergency signaling (SOS), location and last-known-location, movement and fall detection, heart-rate support monitoring, battery health, checkpoint attendance, connection status, geofencing, group-separation detection, and wearable-integrity (removal) detection. Each is analyzed against four questions: what problem it solves, what the organizer actually sees, how it is technically implemented, and what action the organizer can take as a result.

SAFE LINK is explicitly scoped as an organized-group safety tool operating within a defined event area — not a military system, not a medical diagnostic device, and not a general-purpose tracking product. The proposal distinguishes clearly between what a prototype (MVP) can realistically deliver using off-the-shelf embedded components (ESP32/STM32-class microcontrollers, a GNSS module, an IMU, an optical heart-rate sensor, and a low-power radio), what the fuller proposed architecture supports, and what remains future work. Every feature in this document is designed to fail safely: an unresponsive device is reported as unknown, never silently assumed to be fine.

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List of Abbreviations / Acronyms

API	Application Programming Interface
BLE	Bluetooth Low Energy
BOM	Bill of Materials
GNSS	Global Navigation Satellite System
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
HR	Heart Rate
I2C	Inter-Integrated Circuit (communication bus)
IMU	Inertial Measurement Unit
INSA	Information Network Security Administration
IoT	Internet of Things
IP65	Ingress Protection rating 65 (dust-tight, water-jet resistant)
LoRa	Long Range (low-power wide-area radio protocol)
MCU	Microcontroller Unit
MVP	Minimum Viable Product
PPG	Photoplethysmography
REST	Representational State Transfer
SOS	International distress signal / emergency call for help
SPI	Serial Peripheral Interface
TLS	Transport Layer Security
UART	Universal Asynchronous Receiver-Transmitter
WBS	Work Breakdown Structure

CHAPTER ONE

Introduction

1.1 Background of the Study/Project

Organized outdoor activities — youth camps, university excursions, school field trips, hiking and scouting groups, outdoor training programs, and adventure tourism operations — routinely place a small number of adult supervisors in charge of large groups spread across open, visually obstructed, or physically dispersed terrain. This is the operating context in which SAFE LINK is proposed.

Embedded and IoT technology — low-power microcontrollers, GNSS positioning, inertial sensing, and low-power wireless communication — has matured to the point where compact, wrist-worn devices can reliably report a participant's status to a central system throughout an activity, rather than relying solely on periodic manual checks. SAFE LINK applies this technology to the specific, recurring problem of group safety supervision described in Section 1.2.

1.2 Statement of the Problem

Organized outdoor activities routinely place a small number of adult supervisors in charge of dozens or hundreds of participants spread across open, visually obstructed, or physically dispersed terrain. The prevailing tools for keeping track of everyone — headcounts, buddy systems, handheld radios, and scheduled roll calls — are all manual, periodic, and fundamentally reactive: they surface a problem only when a person actively checks and notices something is wrong, often minutes after the underlying event occurred.

This creates several recurring failure modes. A participant who falls or becomes incapacitated on a trail may not be discovered until the next scheduled checkpoint. A child who wanders from their assigned group during free time may go unnoticed until an evening headcount comes up short, at which point the search has to start from the last confirmed sighting rather than a recent position. A medical or safety event that a participant cannot verbally report — because they are alone, injured, or out of radio range — has no path to reach an organizer at all. Even when nothing goes wrong, organizers spend a disproportionate share of their attention on routine verification (“is everyone still here?”) rather than on the exceptions that actually require judgment and intervention.

SAFE LINK is proposed as a response to this pattern: not a replacement for organizer judgment or physical supervision, but a system that continuously performs the low-value, high-frequency task of noticing — so that human attention is reserved for deciding what to do once something has been noticed.

1.3 Purpose of the Project/Research

SAFE LINK closes the loop between participant, wearable, backend system, and organizer dashboard. Each participant wears a compact device that collects safety-relevant signals — location, motion, connectivity, battery, and (as a supporting signal) heart rate — and transmits them to a central system. An alert engine continuously evaluates each participant's state against defined thresholds and raises an alert only when something falls outside the expected pattern. The organizer interacts with a dashboard, not with raw sensor feeds.

1.4 General Objective

> ***To design a wearable-based group safety and participant monitoring system — SAFE LINK — that converts participant sensor telemetry into a small set of exception-based alert states, enabling a small organizing team to responsibly oversee an outdoor group of significant size.***

1.5 Specific Objectives

- To analyze existing manual group-monitoring practices and identify their structural limitations.
- To design a wrist-worn wearable hardware architecture around feasible, off-the-shelf embedded components.
- To design an exception-based alert engine and state machine (SAFE, WARNING, CRITICAL, UNKNOWN) that minimizes false alerts while never silently masking a genuine exception.
- To design an organizer dashboard that surfaces only participants requiring attention, rather than raw continuous telemetry.
- To define a communication and backend architecture connecting wearables, gateways, and the dashboard with graceful degradation on connectivity loss.
- To define a clearly scoped Minimum Viable Product (MVP) achievable within a student team's timeline and skill set.
- To define a staged testing and validation plan that establishes confidence in detection thresholds before field reliance.

1.6 Research Questions

- To what extent can low-cost, off-the-shelf embedded sensors (IMU, GNSS, PPG) reliably distinguish genuine safety exceptions from normal activity noise in an outdoor group setting?
- What combination of signals and thresholds minimizes false-positive alert rates without delaying detection of genuine falls, separations, or SOS events?
- What is an appropriate accountability boundary between automated detection and organizer judgment for each class of alert?
- How should the system behave when connectivity, GNSS visibility, or a sensor reading is unavailable, so that silence is never misread as safety?

1.7 Significance of the Project

- For organizers: fewer manual headcounts and radio checks, materially faster awareness of emergencies, and a system that tells them who needs attention instead of requiring them to work it out.
- For institutions and operators: a documented, timestamped record of monitoring activity and incident response, supporting — though not by itself guaranteeing — a demonstrable duty-of-care process.
- For families and stakeholders: visible evidence that a structured, technology-assisted safety process exists alongside standard supervision practices.
- For participants: a low-friction, unobtrusive device that does not interfere with the activity itself — no app to operate, no continuous action required, only a button to press if something is genuinely wrong.

Underlying all four is the same structural shift: from continuously watching everyone, to being told when someone deviates from the expected pattern. This is what distinguishes SAFE LINK from "a collection of sensors connected to a website" — every hardware signal in the system exists because it feeds a specific, named organizer decision, not because it was easy to measure.

1.8 Scope of the Project

This proposal is scoped to organized-group safety and accountability monitoring within a defined event or activity area, covering identification, SOS signaling, location and last-known-location, movement and fall detection, heart-rate support monitoring, battery health, checkpoint attendance, connection status, geofencing, group-separation detection, and wearable-integrity detection, together with the wearable hardware, communication, backend, and dashboard architecture needed to support them.

The project explicitly does not cover: military-grade tracking or anti-jamming capability; medical diagnostic functionality of any kind; general-purpose global tracking outside a defined event; or guaranteed operation in every possible venue or connectivity condition. What the MVP prototype delivers, versus what remains future work, is drawn as a clear line in Table 7 (Section 3.6 / Chapter Three).

1.9 Limitations

- Technical: GNSS accuracy degrades under canopy or indoors; optical heart-rate sensing degrades under motion; low-cost IMU heuristics carry a non-zero false-positive rate (Chapter Four, Section 4.5).
- Financial: component and prototyping costs are illustrative only and should be validated against current supplier quotes before any build-out (Chapter Six).
- Time: the proposal is scoped to a summer-camp timeline, favoring staged subsystem validation over building a fully integrated system first (Chapter Five).
- Resource: the prototype is designed around a student team's realistic skill set — embedded firmware, basic backend/web development, and basic hardware assembly — rather than custom silicon or specialized fabrication.

1.10 Definition of Key Terms

- SAFE / WARNING / CRITICAL / UNKNOWN — the four operational states the alert engine assigns to each participant; UNKNOWN is a distinct connectivity state, never conflated with SAFE.
- MVP (Minimum Viable Product) — the reduced-scope prototype deliverable, distinguished throughout this proposal from future enhancements.
- Geofence — a defined circular or polygonal boundary that triggers an alert when a participant's location fix falls outside it.
- Centroid (group) — the mean position of a group's currently-online members, used as the reference point for separation detection.
- Heartbeat packet — a lightweight periodic transmission confirming a wearable is alive and in range, independent of full telemetry.
- Exception-based monitoring — the governing design philosophy of surfacing only state changes that require organizer attention, rather than continuous raw data.

CHAPTER TWO

Literature Review

2.1 Introduction

This chapter situates SAFE LINK against current manual group-monitoring practice and the general landscape of embedded sensing technology, and identifies the specific gap the proposed system is designed to close.

2.2 Theoretical Background

The design rests on two established engineering principles rather than a single novel technique. The first is exception-based (or "management by exception") monitoring: a supervisory system should surface only state changes that require a decision, rather than requiring a human to continuously interpret raw data — a principle widely used in industrial SCADA and alarm-management systems and adapted here to a small-team outdoor-safety context. The second is multi-signal sensor fusion: combining several individually noisy, low-cost signals (acceleration magnitude, orientation change, and stillness duration, for example) into a single classification is more reliable than trusting any one signal in isolation, which is why the system frames automated detections as "possible" events requiring verification rather than confirmed diagnoses.

2.3 Existing Systems

Current practice across nearly all the use cases this proposal targets relies on some combination of headcounts, buddy systems, handheld radios, and verbal roll calls. Each is examined below for what it tells an organizer and, more importantly, what it structurally misses.

Current practice across nearly all of the use cases above relies on some combination of headcounts, buddy systems, handheld radios, and verbal roll calls. Each of these is a reasonable tool in isolation, but each has a structural limitation that SAFE LINK is designed to address without discarding the underlying practice — the system is intended to sit alongside these methods, not replace organizer judgment.

Method	What It Tells You	What It Misses
Headcount	Whether the expected number of people is present, at the moment counted	Who specifically is missing, where they were last seen, and how long they have been unaccounted for
Buddy system	Whether a peer has recently seen the participant	Depends on peer attentiveness and honesty; breaks down for younger participants or large groups
Handheld radio	Real-time voice contact with whoever is actively listening and near a radio	Does not scale past a few group leaders; requires someone to notice a problem and actively report it
Scheduled roll call	A confirmed, named list of who is present at that instant	Nothing between roll calls; time-consuming and disruptive if run frequently enough to close that gap

None of these methods are continuous, and none of them automatically surface time since last confirmed status — a quantity that matters enormously in an emergency but that no manual method tracks by default. SAFE LINK does not ask organizers to trust a machine over their own judgment; it removes the need to manually poll every participant in order to find out who, if anyone, currently needs that judgment applied.

2.4 Review of Related Studies

Consumer fitness trackers and standalone GPS trackers address adjacent but distinct problems. Fitness wearables optimize for personal health metrics over days or weeks, not real-time group accountability during a bounded event; standalone GPS trackers report position but generally lack fall detection, wearable-integrity sensing, or any concept of "group" position. Neither category is designed around an organizer-facing exception dashboard, a defined event geofence, or a documented alert-acknowledgment workflow — the specific requirements that recur across the use cases identified in Chapter One.

The system is designed to generalize across several categories of organized outdoor group, each of which shares the same underlying structure — one or a few responsible adults, many participants, an open or semi-open area — but differs in group size, participant age, activity duration, and terrain.

User Type	Representative Scenario	Primary Need
Youth camps	Week-long overnight camp, ages 8–14, multiple activity zones	Continuous headcount and fast SOS/fall response for minors who may not self-report
University outdoor clubs	Weekend hiking or climbing trip, small self-organized groups	Separation detection on trail; reliable last-known-location if signal drops
School excursions	Day trip to a museum, park, or historical site with 100+ students	Rapid check-in and attendance verification at checkpoints
Scouting activities	Multi-day wilderness trip with staged campsites	Per-site geofencing; battery life across a multi-day itinerary
Field trips	Educational site visits with a fixed itinerary and volunteer chaperones	Simple, low-training-overhead operation
Outdoor training programs	Physically demanding courses — obstacle courses, endurance training	Fall and prolonged-inactivity detection; heart rate as a supporting signal
Adventure tourism operators	Paid guided excursions — rafting, ziplining, guided trekking	SOS response and a defensible accountability / incident record

Despite their differences, every one of these contexts reduces to the same operational question that motivates the system's design: at any given moment, which participants require attention, and what is the fastest path from "something is wrong" to "an organizer knows and can act"?

2.5 Existing Technology

The component classes proposed for SAFE LINK — ESP32-class microcontrollers, u-blox NEO-M8N–class GNSS receivers, MPU6050-class inertial measurement units, MAX30102-class optical heart-rate sensors, and MAX17048-class fuel gauges — are mature, widely documented, hobbyist-to-professional parts (see Table 4, Chapter Three). Their maturity is precisely what makes a staged, evidence-based MVP feasible on a compressed student timeline, as opposed to requiring custom silicon or unproven sensing approaches.

2.6 Research/Technology Gap

Existing manual methods (Section 2.3) are periodic and reactive; existing consumer wearable technology (Section 2.4) is individually capable but not organized around group accountability. The gap this proposal targets sits at their

intersection: a system that (a) continuously classifies each participant into a small set of operationally meaningful states, (b) treats connectivity loss as an explicit UNKNOWN state rather than silently defaulting to “safe,” and (c) is scoped specifically to the organizer-decision workflow of a bounded outdoor event — registration, checkpoints, geofencing, and group-separation detection — rather than to general-purpose personal tracking. SAFE LINK's eleven-feature design (Chapter Three) is proposed as a direct response to this gap.

2.7 Conceptual Framework

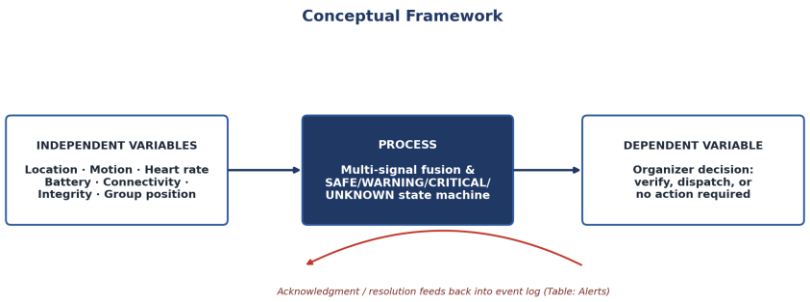


Figure 1. Conceptual framework — independent variables, process, and dependent variable.

Sensor-derived independent variables (location, motion, heart rate, battery, connectivity, wearable integrity, and group position) are fused by the alert engine's state-machine process into a single dependent variable: the decision an organizer is asked to make. Acknowledged and resolved alerts feed back into the event log, so the framework is a closed loop rather than a one-way pipeline — a resolved alert is retained as a record, never silently discarded.

2.8 Summary of Literature Review

Manual monitoring methods and general-purpose consumer wearables each solve part of the group-safety problem but not the whole of it. SAFE LINK's contribution is architectural rather than a claim of novel sensing hardware: an exception-based, fail-safe state machine purpose-built for the organizer-decision workflow of a bounded outdoor event, built from technology that is already well understood and readily available.

CHAPTER THREE

Methodology

3.1 Introduction

This chapter explains how SAFE LINK will be developed: the overall approach, the functional and non-functional requirements the system must satisfy, the system's hardware and software design, component selection, implementation flow, testing plan, data-collection approach, and ethical considerations.

3.2 Research/Development Approach

The project follows a staged prototyping approach rather than a single-pass build: individual subsystems (GNSS fix acquisition, IMU threshold tuning, HR signal quality, battery discharge calibration) are proven independently on breadboards or development boards first, then integrated onto a single custom PCB once subsystem-level behavior is understood, and only then fitted into an enclosure.

The recommended prototyping path is staged: individual subsystems proven on breadboards or off-the-shelf development boards first (GNSS fix acquisition, IMU threshold tuning, HR signal quality, battery discharge calibration, each independently), then integration onto a single custom PCB once the subsystem-level behavior is understood, and only then a fitted enclosure. Building the full integrated board first, before any subsystem has been individually validated, is the most common way this kind of prototype loses time to debugging — the staged approach is recommended specifically to avoid that.

Fall detection accuracy, separation-detection thresholds, heart-rate signal quality under motion, and integrity-loop durability under repeated flexing are flagged here explicitly as the features most likely to need real field data before their thresholds can be trusted — a point developed further in Sections 21 and 24.

3.3 Requirements Analysis

3.3.1 Functional Requirements

Each of the eleven core features below is analyzed against the same four questions: the problem it solves, what the organizer sees, how it is technically implemented, and what action the organizer can take as a result. Dashboard examples use a consistent illustrative participant (Participant 023, device WG-023, Hiking Team B) throughout for continuity.

Feature 1 — Participant Identification

Purpose

Every other feature in this system is only useful if an incoming signal can be traced to a specific person. Identification is the foundation the rest of the system is built on: it is what turns “a device reported a fall” into “Participant 023 may have fallen.”

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — IDENTIFICATION CARD

```
Participant: Participant 023
Device ID: WG-023
```

```
Group: Hiking Team B
Status: NORMAL
Last Update: 10:42:16
```

Technical Implementation

Each wearable carries a unique device identifier — either derived from the microcontroller's factory-programmed hardware ID (for example, the MAC-derived identifier available on ESP32-class chips) or a separately assigned serial number written to flash during manufacturing or setup. During pre-event registration, an organizer links a physical device to a participant record in the database, ideally by scanning a QR code printed on the wearable's strap with a registration tablet or phone rather than typing IDs manually. Every telemetry and alert packet the device sends carries this device ID; the backend performs the participant lookup, meaning identity is a database-side association rather than something the device itself needs to know or compute.

This mechanism has a clear limitation, stated explicitly rather than glossed over: it proves which device transmitted the data, not that the correct person is physically wearing it. That gap is intentional — it is the reason Feature 11 (Wearable Integrity and Removal Detection) exists as a complementary control rather than being folded into identification itself.

Organizer Response

Organizers can reassign a device to a different participant record if devices are swapped, and the registration workflow flags any device transmitting without a matching participant assignment before the event begins.

Feature 2 — SOS Button

Purpose

Automatic detection cannot catch everything — a participant who feels unsafe, is lost but not yet flagged as separated, or is experiencing a non-fall medical issue needs a direct way to escalate. The SOS button is the participant's manual override on top of all the automatic monitoring described elsewhere in this document.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — CRITICAL SOS ALERT

```
⚠ CRITICAL SOS ALERT
Participant: Participant 023
Device: WG-023
Group: Hiking Team B
Location: [Current or Last Known Coordinates]
Time: 10:42:16
Status: ASSISTANCE REQUIRED
```

Technical Implementation

A dedicated physical button requires a deliberate long-press (on the order of one to two seconds) rather than a single tap, specifically to reduce accidental activation from a brushing branch or a shifted pack. On activation, the microcontroller immediately assembles a high-priority packet — bypassing its normal duty-cycle sleep schedule — tagged with an SOS flag, the current or most recent GPS fix, a timestamp, and battery status, and writes the event to

local flash before attempting transmission, so the event survives even if the radio link fails at that instant. A short haptic pulse and LED flash confirm to the participant that the press registered, which matters because the device has no screen to otherwise show feedback.

The device retransmits at short intervals until it receives an application-layer acknowledgment from the backend — not merely a radio-layer ACK from the gateway — because delivery to a gateway is not the same thing as an organizer having seen the alert. The device gives different feedback once acknowledgment is confirmed versus while still retrying. A short cancel window (releasing before the hold threshold, or a brief confirmation buzz with a follow-up cancel option) keeps a false trigger recoverable without requiring the participant to explain themselves after the fact.

Organizer Response

The dashboard visually prioritizes SOS above all other alert types and requires explicit organizer acknowledgment, recording who acknowledged it and when. The alert remains active on the dashboard — not auto-cleared — until the organizer logs a resolution.

Feature 3 — Location and Last Known Location

Purpose

Provide a participant's position when it is needed, and — just as importantly — preserve and clearly label the last successfully received position when live tracking is temporarily unavailable, so organizers are informed rather than left blind.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — LIVE POSITION

```
Current Location: 9.0357, 38.7651
Last Updated: 12 seconds ago
Location Status: CURRENT
```

ORGANIZER DASHBOARD VIEW — SIGNAL LOST

```
LOCATION STATUS: LAST KNOWN LOCATION
Last Position: 9.0357, 38.7651
Last Received: 14 minutes ago
```

Technical Implementation

A GNSS receiver (a u-blox NEO-M8N–class module is a reasonable MVP choice) acquires a position fix at a configurable interval. Because continuous high-rate GPS acquisition is one of the largest single power draws in the wearable, the MVP uses an adaptive schedule rather than a fixed timer: fixes are taken more frequently — roughly every 15 to 30 seconds — while the device's own activity classifier (Feature 4) reports the MOVING state, and much less frequently — every several minutes — while STATIONARY, with the accelerometer's motion interrupt able to wake the GNSS early if movement resumes. Each fix is timestamped and attached to the regular telemetry packet. The backend independently stores the most recent valid fix per participant regardless of whether the latest packet contained a fresh one, so “last known location” persists automatically across any gap in reception.

Limitations are stated explicitly rather than assumed away: GNSS requires reasonable sky visibility, and dense forest canopy, canyons, and indoor structures will degrade or block a fix entirely; a cold-start acquisition can take significantly longer than a warm one in poor conditions. The dashboard is required to show a visible freshness indicator and must never silently reuse a stale point as though it were current — CURRENT and LAST KNOWN LOCATION are always distinct, clearly labeled states.

Organizer Response

When a participant shows LAST KNOWN rather than CURRENT, organizers treat the coordinate as a search starting point, not a live position, and can task nearby group members or staff to re-establish physical contact.

Feature 4 — Movement and Activity State Detection

Purpose

Raw accelerometer data is meaningless to an organizer glancing at a dashboard. This feature's job is to convert that raw signal into a small set of operational states an organizer can act on without any technical interpretation.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — NORMAL ACTIVITY

```
Activity Status: MOVING
```

ORGANIZER DASHBOARD VIEW — INACTIVITY WARNING

```
WARNING: PROLONGED INACTIVITY
Duration: 18 minutes
Last Known Location: [Coordinates]
```

Technical Implementation

A 6-axis inertial measurement unit (an MPU6050-class accelerometer/gyroscope combination is a common, well-documented MVP choice) is sampled continuously by the microcontroller. Firmware computes rolling features over a short window — typically the magnitude and variance of acceleration over roughly thirty to sixty seconds — to classify MOVING versus STATIONARY, and raises PROLONGED INACTIVITY once the STATIONARY state persists beyond an activity-appropriate threshold. That threshold must be tunable per activity type: a short window suits an active hike, while a longer one is needed for a scheduled rest stop or overnight sleep period, where flagging every quiet moment would make the alert meaningless.

POSSIBLE FALL detection specifically requires a combination of signals: a short high-magnitude acceleration spike consistent with an impact, together with a rapid orientation change, followed by a subsequent stillness window. Requiring all three in sequence is what distinguishes a fall pattern from an ordinary jump, a fast sit-down, or a participant simply running — each of which can trigger one of the three signals in isolation. This is explicitly framed as a heuristic, not a confirmed diagnosis, which is why the system reports it as Possible Fall Event and always routes it to organizer verification rather than auto-escalating it as a confirmed injury.

Organizer Response

On a possible fall alert, the recommended first step is attempting radio or verbal contact through nearby group members before dispatching a response, since low-cost IMU-based heuristics are expected to produce a meaningful false-positive rate; if contact fails within a short window, the alert follows the same verify-then-escalate pattern as SOS.

Feature 5 — Heart Rate Monitoring

Purpose

Heart rate is included strictly as a supporting physiological signal — never as a medical diagnosis and never as proof that a participant is or is not alive. Framing this correctly is a safety requirement in its own right, not just a legal caveat.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — NORMAL RANGE

```
Heart Rate: 78 BPM
Signal Quality: GOOD
Status: NORMAL RANGE
```

ORGANIZER DASHBOARD VIEW — READING UNAVAILABLE

```
HEART RATE ALERT
Participant: Participant 023
Reading: Unavailable / Abnormal
Requires: Manual Verification
```

Technical Implementation

A reflective optical (PPG) sensor such as a MAX30102-class module reads pulse signal against wrist skin over I2C. To conserve power, the device takes a short measurement window every few minutes rather than sampling continuously, since continuous PPG capture with motion-artifact filtering is one of the more power- and processing-intensive subsystems in the wearable. Firmware and backend both apply basic signal-quality checks — using perfusion index or signal-amplitude thresholds — before publishing a reading, and report SIGNAL POOR or UNAVAILABLE rather than fabricating a number when skin contact or motion conditions are inadequate.

Wrist-worn optical PPG is known to degrade substantially under motion, which is precisely when a reading might matter most — this is a general limitation of the sensing approach, not specific to any one component choice, and the proposal does not claim otherwise. A missing or abnormal reading is routed to “manual verification required,” never to an automated medical-emergency or fatality inference; this distinction protects against a serious failure mode in which the system would otherwise appear to claim something a $10 optical sensor and a loose strap cannot actually establish.

Organizer Response

Abnormal or missing heart-rate data is treated as one input among several — cross-referenced against recent movement, connection status, and whether an SOS has been raised — rather than acted on in isolation.

Feature 6 — Battery Monitoring

Purpose

A wearable with a dead battery looks, from the organizer's point of view, identical to a participant who is simply fine and has nothing to report — unless the system proactively flags the battery state before that happens.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — NORMAL BATTERY

```
Battery: 82%
Status: NORMAL
```

ORGANIZER DASHBOARD VIEW — LOW BATTERY

```
LOW BATTERY
Participant: Participant 023
Battery: 15%
Recommended Action: Recharge or replace device
```

Technical Implementation

State of charge is best measured with a dedicated fuel-gauge IC (a MAX17048-class coulomb counter is a common, low-cost MVP choice) rather than a bare voltage divider, because Li-ion cell voltage sags noticeably under load — particularly during a GPS or radio transmission current spike — which makes a simple voltage-to-percentage lookup noisy. If a fuel-gauge IC is not used, a voltage-divider approach is still workable but must be calibrated against a discharge curve specific to the chosen cell chemistry and capacity. Charging state is read directly from the charge-management IC's status output. Threshold alerts (for example, a 20% low-battery warning and a 5% critical warning) are computed from the reported percentage, either on-device or backend-side.

Organizer Response

A LOW BATTERY warning is routine triage — plan a swap or recharge at the next convenient stop. A CRITICAL battery warning is treated differently: it is a prompt to proactively check that participant's status while there is still time to do so, converting a battery warning into a location and status check rather than waiting for the device to go dark.

Feature 7 — Check-in / Attendance Status

Purpose

Verify that participants are accounted for at defined moments — an assembly, a meal, a return to camp — as an event-accountability tool, not a continuous surveillance feed and not a generic school-attendance system.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — CHECKPOINT SESSION

```
MORNING ASSEMBLY
Expected: 120
Checked In: 116
```

```
Missing: 4

Participant 023 – Last Known Location: Dormitory Area
Participant 041 – Device Offline
```

Technical Implementation

For the MVP, proximity-based BLE check-in is the recommended mechanism: a fixed BLE beacon or gateway is placed at the checkpoint, and wearables within range are automatically registered as present when an organizer opens a checkpoint session on the dashboard or a tablet, matched against the full expected roster so the system can name who is missing rather than only report a count. This is preferred over the alternatives for the MVP specifically because it requires no action from the participant — NFC or RFID check-in is more precise but requires each participant to individually tap a reader, which becomes a bottleneck and a burden for younger children in a large group; QR-code organizer-scanned check-in is a reasonable manual fallback but adds staff time per participant. Proximity detection trades a small amount of boundary ambiguity — mitigated with a minimum dwell time or signal-strength threshold rather than a single ping — for check-in that happens automatically as the group gathers.

Organizer Response

The dashboard surfaces missing participants by name with their last known location, so follow-up begins immediately rather than requiring a manual cross-reference against a paper roster.

Feature 8 — Connection Status

Purpose

This feature exists to protect a single, critical distinction: confirmed safe is not the same thing as we have not heard otherwise. Treating silence as safety is a design error the system is built to avoid throughout.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — HEALTHY CONNECTION

```
Connection: ONLINE
Last Communication: 8 seconds ago
Signal: GOOD
```

ORGANIZER DASHBOARD VIEW — CONNECTION LOST

```
STATUS: UNKNOWN – CONNECTION LOST
Last Contact: 10 minutes ago
Last Known Location: [Coordinates]
```

Technical Implementation

Lightweight heartbeat packets, sent independently of full telemetry, confirm a device is alive and in range even when it has nothing else to report. The backend tracks a last-successful-contact timestamp per device and applies a timeout — typically two to three times the expected heartbeat interval, to tolerate normal jitter and retries — before transitioning a participant through WEAK CONNECTION, DELAYED, and finally OFFLINE / STATUS UNKNOWN.

The governing design rule is stated plainly because it is easy to get wrong: the absence of an alert must never be treated as equivalent to a confirmed-safe status. The dashboard actively distinguishes a state that was recently confirmed from one that has simply gone quiet, rather than defaulting an unresponsive device to look identical to a healthy one.

Organizer Response

STATUS UNKNOWN triggers a manual-check task — radio the nearest group leader or dispatch someone — distinct from, and generally less urgent than, a CRITICAL alert, but never allowed to silently age out of view unresolved.

Feature 9 — Geofencing

Purpose

Convert “the area this activity is expected to stay within” into an automatic boundary check, instead of relying on a group leader's visual judgment across an entire group's footprint.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — GEOFENCE ALERT

```
GEOFENCE ALERT
Participant: Participant 023
Event: Exited Safe Zone
Time: 10:42:16
Location: [Coordinates]
```

Technical Implementation

Zones are defined as either a circle (center point plus radius — the simplest option, well suited to a single basecamp) or a polygon (an ordered list of coordinates — better suited to trail corridors or irregular property boundaries). The MVP recommendation is to evaluate zone membership backend-side against each incoming location fix rather than on-device: this centralizes the logic, allows zones to be edited mid-event without re-provisioning every wearable, and is simpler to implement and test on a compressed prototype timeline. On-device evaluation — which reduces alert latency and backend load at the cost of needing zone definitions synced to every wearable — is noted as a later optimization rather than an MVP requirement. The system should support multiple concurrent zones, including nested safe and restricted areas, with independently configurable alert severity.

Organizer Response

On exit, organizers see who left, from where, and how long ago, and can visually confirm on the map whether it reflects a genuine excursion or a GPS fix jittering right at a boundary edge before deciding how to respond.

Feature 10 — Grouping and Separation Detection

Purpose

Catch a participant drifting away from their assigned group before they become fully lost — where “lost” here is defined relative to the group's own position, not a fixed map boundary.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — SEPARATION ALERT

```
GROUP B
Members: 12
Within Expected Area: 11
Potentially Separated: 1

Participant: Participant 023
Distance from Group: 1.1 km
```

Technical Implementation

The specification explicitly rules out vague language such as “far away,” so this feature requires a concrete, defined metric. The MVP recommendation is distance from the group centroid — the mean position of all currently-online members of the group — rather than distance from a designated leader (not every group structure names one, and a leader's device is a single point of failure for the whole metric) or distance from the nearest member (cheap to compute, but can mask an entire sub-group of two or three people drifting together, and is noisy for small groups).

The centroid approach has its own known weakness, stated directly: a single member who is already far away pulls the centroid toward themselves, potentially masking their own separation. The mitigation is to compute the centroid from members within a normal spread of one another — for example, excluding outliers beyond roughly 1.5 times the median pairwise distance — before measuring anyone's distance from it. Thresholds must vary by activity type: a museum group might reasonably use 30–50 meters, while a hiking trail might allow several hundred meters of normal spacing along the trail itself. A distance threshold is always combined with a time threshold (“beyond X meters for more than Y minutes”) specifically to avoid flagging the brief, normal spreading that happens when someone stops to tie a shoe or the trail bends.

Organizer Response

Organizers first check whether the flagged participant is together with any other named individual — a pair of stragglers reads very differently from a genuine solo separation — and radio the nearest group member before independently investigating.

Feature 11 — Wearable Integrity and Removal Detection

Purpose

Every feature above is only meaningful while the device is actually being worn. This feature closes that gap by detecting when the wearable comes off.

Organizer Dashboard

ORGANIZER DASHBOARD VIEW — INTEGRITY ALERT

```
WEARABLE INTEGRITY ALERT
Participant: Participant 023
Event: DEVICE REMOVED OR TAMPERED
Time: 10:42:16
Last Location: [Coordinates]
Requires: Participant Verification
```

Technical Implementation

The recommended MVP mechanism is a closed conductive loop running through the strap's clasp: while fastened, the loop is closed and a GPIO pin reads continuity; opening the clasp breaks the loop and is detected immediately in firmware. This is favored for a student prototype over capacitive or optical skin-contact sensing — a fitness-tracker-grade approach that is more sensitive to strap tightness, skin tone, and moisture, and would require calibration effort better spent elsewhere on a compressed timeline. A magnetic reed-switch clasp and a mechanical tamper switch are viable alternatives with different trade-offs: a reed switch is very simple but only detects clasp state (not, for example, a cut strap), while a mechanical switch is robust but adds a moving part.

The specification requires distinguishing accidental detachment, legitimate removal (charging, bathing, a medical exam), and possible tampering — a distinction that hardware alone cannot fully resolve. The proposed approach combines the raw removal signal with context: whether it occurred during a scheduled charging window, at a designated rest area, or unexpectedly mid-activity, to set alert severity, while keeping the alert an accountability item requiring verification, never an automatic emergency.

Organizer Response

Locate the participant, confirm status, and either have them re-wear the device — closing the loop clears the alert automatically — or log a supervised exception (for example, “removed for medical exam, participant confirmed present”).

3.3.2 Non-Functional Requirements

- Reliability / fail-safe behavior — an unresponsive device or sensor is always reported as UNKNOWN or UNAVAILABLE, never silently defaulted to SAFE.
- Security — all wearable-to-server transmissions are encrypted (AES-128 floor, TLS on the gateway-to-backend hop); dashboard access is role-based and limited to registered organizer accounts.
- Performance — SOS and possible-fall events bypass normal duty-cycle scheduling and batching for prioritized, retry-until-acknowledged delivery.
- Usability — the organizer dashboard is designed around decisions, not raw sensor values, so non-technical staff can act on it under pressure.
- Scalability — the backend's ingestion and API layers have a defined horizontal-scaling upgrade path, though this is out of scope for the MVP.
- Maintainability — zone, threshold, and severity configuration are backend/database-side wherever feasible, so they can be adjusted without re-provisioning wearables.

3.4 System Design

3.4.1 Overall System Architecture

The organizer dashboard is designed around decisions, not raw sensor values. Every element on screen exists to answer “who needs my attention, and what should I do about it” as directly as possible.

Summary Bar

The top of the dashboard always shows five counts: Total Participants, Safe/Normal, Warning, Critical, and Offline/Unknown. This is the single glance that tells an organizer whether anything currently needs attention before they look at anything else.

Participant List

Name	Group	Status	Battery	Connection	Last Update
Participant 023	Hiking Team B	NORMAL	82%	ONLINE	8s ago
Participant 041	Hiking Team B	UNKNOWN	61%	OFFLINE	10m ago
Participant 017	Cabin Group 3	WARNING	15%	ONLINE	22s ago

The list is sortable and filterable by status, so a busy organizer can immediately narrow to anyone not in the NORMAL state rather than scanning the full roster.

Interactive Map

The map shows live participant positions, defined safe and restricted zones, active alert locations, and last-known positions for anyone currently offline — all layered on a single view so spatial context (who is near whom, who is near a boundary) is immediately visible rather than requiring separate lookups.

Alert Center

A chronological, filterable feed of every alert: type, participant, severity, time, location, acknowledgment status, and a recommended next action. Critical alerts remain pinned at the top until acknowledged and resolved.

Participant Detail Page

Selecting any participant opens a full profile: identification, assigned group, current and last-known location, activity state, heart rate, battery, connection status, wearable integrity status, recent alert history, and a historical event timeline — everything needed to make a decision about that one person without navigating away.

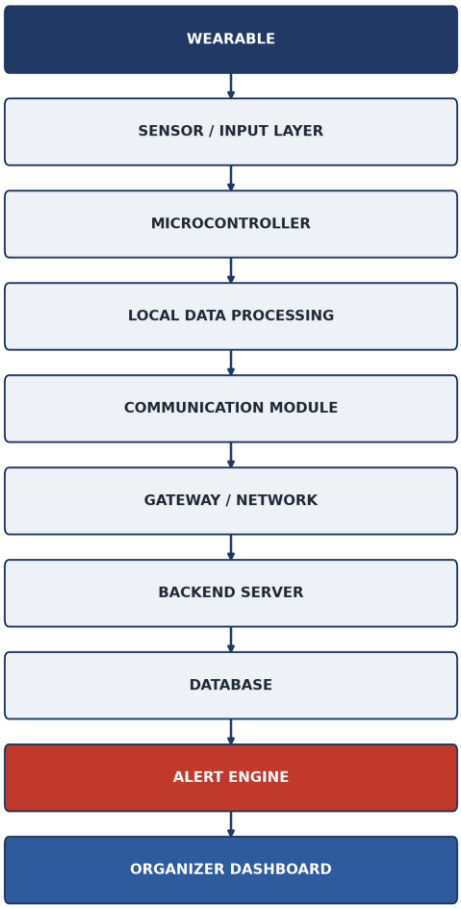


Figure 2. End-to-end system architecture, from wearable sensors to the organizer dashboard.

Each layer maps directly to a section of this proposal: the sensor and microcontroller layers are detailed in Sections 9–10, local processing (on-device classification and buffering) is described throughout Section 7, the communication and gateway layers in Section 11, the backend, database, and alert engine in Sections 12–14, and the dashboard in Section 8. The chain is designed so that a failure at any single layer degrades gracefully rather than silently — a theme developed further in Section 22.

3.4.2 Hardware Design

A wrist-worn form factor is recommended over a chest strap or clip-on badge. Several features independently point the same direction: the optical heart-rate sensor (Feature 5) requires sustained skin contact best achieved at the wrist; the removal-detection loop (Feature 11) maps naturally onto a strap clasp; and a wristband is the most tolerant form factor for participants of varying ages and activity types. The enclosure should target a reasonable degree of dust and water resistance (an IP65-class rating is a sensible design goal) given outdoor use in variable weather, and should be low-profile enough not to interfere with normal activity. The SOS button should be large enough to operate without fine motor precision, while remaining recessed or requiring a deliberate long-press so it cannot be triggered by an accidental brush.

Wearable Internal Architecture

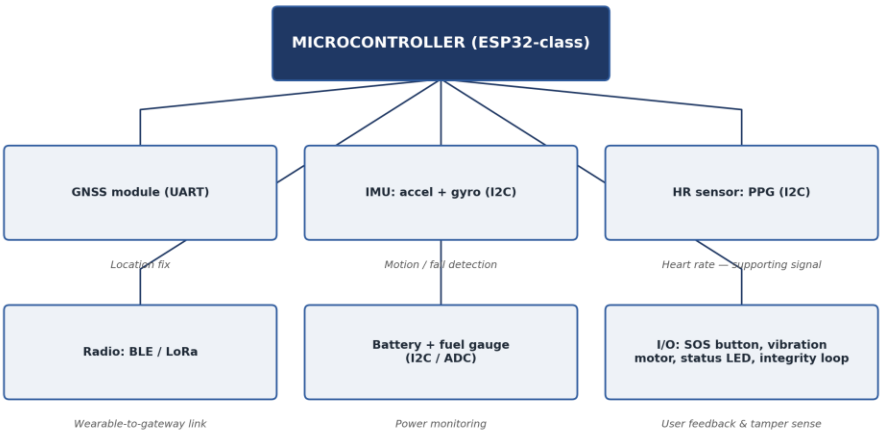


Figure 3. Wearable internal architecture — the ESP32-class microcontroller as the single coordination point for all onboard sensors and I/O.

The microcontroller is the single coordination point: it schedules sensor sampling, runs the on-device activity classifier, buffers events to local flash, manages the adaptive GNSS/HR sampling rates described in Sections 7 and 10, and drives the radio link. Everything else in this document — the alert logic, the dashboard, the communication architecture — is built on top of what this one chip reliably reports.

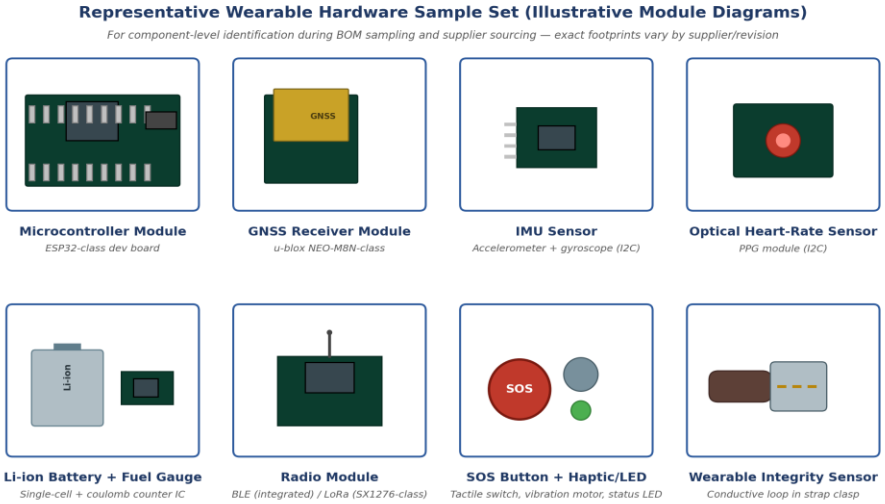


Figure 4. Representative wearable hardware sample set (illustrative module diagrams) — for component-level identification during BOM sampling and supplier sourcing; exact footprints vary by supplier and revision.

3.4.3 Software Design

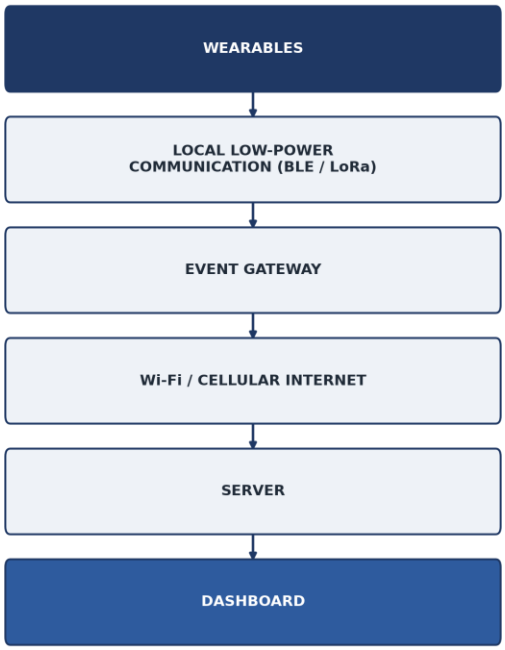


Figure 5. Communication architecture — from wearable to organizer dashboard.

The wearable communication layer connects each device to a nearby gateway using a short-range, low-power link. BLE is the recommended MVP default: it is simple, low-cost, and adequate for a compact, single-campsite footprint, with a typical open-air range on the order of tens of meters. For deployments spread across a trail or a larger property where gateway placement is necessarily sparser, LoRa is the recommended upgrade path — a lower data-rate, longer-range link (realistically hundreds of meters to a few kilometers depending heavily on terrain and antenna placement, not a fixed guaranteed figure) better suited to that footprint. The choice should be made per deployment rather than assumed universally.

The gateway aggregates telemetry from every wearable in range and forwards it over a Wi-Fi or cellular backhaul to the cloud backend. Because backhaul availability cannot be assumed at every outdoor venue, the gateway buffers incoming data locally during any Internet outage and synchronizes once connectivity returns. This is deliberately a second layer of resilience on top of the wearable's own local event buffer (Section 7, SOS handling) — a temporary loss of the gateway's Internet connection does not mean data is lost, only delayed.

The proposal makes no claim of military-grade anti-jamming capability, satellite connectivity, or guaranteed operation in any environment; backhaul availability is deployment-specific and should be surveyed at each venue before an event, with the manual fallback procedures in Section 22 available as a matter of course rather than an exception.

The backend consists of four logical components: an ingestion endpoint that receives packets from gateways, an alert engine that evaluates each participant's state against the thresholds described in Section 14, a data API that serves the dashboard (a REST interface for queries plus a push channel — WebSocket or equivalent — so the dashboard reflects new alerts immediately rather than polling), and an authentication layer restricting dashboard access to registered organizer accounts.

For a student prototype, a single small cloud instance with a managed database is sufficient: expected load is on the order of tens to a few hundred devices reporting on a multi-second to multi-minute cadence, not a scale that justifies

a distributed microservice architecture or a managed message-queue cluster. Recommending that level of infrastructure for this stage would be poor engineering advice — it adds operational complexity without a corresponding benefit at MVP scale. A clear upgrade path (horizontal scaling of the ingestion and API layers, a managed queue between ingestion and the alert engine) exists if the system is later deployed across many simultaneous events, but it is explicitly out of scope for the prototype.

Table	Key Fields	Purpose
Participants	participant_id, name, group_id	Roster and identity records
Devices	device_id, assigned_participant_id, last_seen	Device-to-participant mapping (Feature 1)
Groups	group_id, name, member_ids	Grouping for separation detection (Feature 10)
Geofences	zone_id, type (circle/polygon), coordinates, severity	Zone definitions (Feature 9)
Checkpoints	checkpoint_id, location, session_time	Check-in session records (Feature 7)
LatestState	participant_id, location, battery, hr, activity_state, connection_state	Current status per participant — what the dashboard reads directly
Alerts	alert_id, participant_id, type, severity, timestamp, acknowledged_by, resolution_notes	Full alert and response history
OrganizerUsers	user_id, name, role, access_level	Dashboard authentication and role-based access

Incoming telemetry updates the LatestState table directly — this is the record the dashboard reads, so the dashboard is always looking at current state rather than replaying a log. The alert engine evaluates each update against the rules in Section 14 and writes to the Alerts table only when a participant's state actually changes; a continuously-SAFE participant never produces an Alerts row. This is a deliberate data-minimization choice, consistent with the privacy principles in Section 23: the system stores current state and a record of exceptions, not a continuous raw location log.

Severity	Triggers	Organizer Action Required
Critical	SOS activated; possible fall combined with prolonged inactivity; other high-confidence emergency combinations	Immediate acknowledgment and response
Warning	Geofence violation; low battery; prolonged inactivity; potential group separation; wearable removal	Verification and follow-up, not necessarily immediate dispatch
Information	Normal check-in; movement status changes; charging; group assignment	No action — visible in history, collapsed from the main alert view

This hierarchy exists specifically to prevent alert overload: without it, a dashboard reporting every state change with equal visual weight would train organizers to tune out alerts altogether, which is worse than not having the system at all. Representative decision rules, simplified for illustration:

```
IF impact_magnitude > IMPACT_THRESHOLD
AND orientation_change > ORIENT_THRESHOLD
AND subsequent_stillness_duration > STILL_WINDOW
THEN raise "POSSIBLE_FALL" (CRITICAL, requires verification)

IF distance(participant, group_reference) > SEPARATION_DISTANCE
AND duration_exceeded > SEPARATION_TIME
THEN raise "POSSIBLE_SEPARATION" (WARNING)
```

```
IF time_since_last_contact > 3 * HEARTBEAT_INTERVAL
THEN set status -> "UNKNOWN" (WARNING — not SAFE)
```

Each participant is conceptually a small state machine (SAFE → WARNING → CRITICAL, with UNKNOWN as an orthogonal connectivity state). Anything reaching WARNING or above requires explicit organizer acknowledgment to clear — the system does not silently auto-resolve an alert once the underlying condition passes, since a state that resolved itself without anyone confirming why is exactly the kind of event that should be logged and reviewed, not erased.

3.5 Component/Technology Selection

All components below are widely available, well-documented hobbyist-to-professional parts with mature software libraries — a deliberate choice to keep the prototype achievable on a student team's timeline rather than requiring custom silicon or exotic sourcing.

Component	Example Part	Interface	Role
Microcontroller	ESP32 (or STM32 + external radio)	—	Central processing, sensor scheduling, on-device classification, radio control. ESP32 is recommended for the MVP because its integrated Wi-Fi/BLE removes the need for a separate radio module.
GNSS receiver	u-blox NEO-6M / NEO-M8N	UART (NMEA)	Position fix acquisition for Feature 3. NEO-M8N supports multi-constellation reception for improved accuracy over NEO-6M.
IMU	MPU6050 (accelerometer + gyroscope)	I2C	Motion classification, fall-pattern detection (Feature 4).
Heart-rate / PPG sensor	MAX30102	I2C	Supporting physiological signal (Feature 5).
Battery + charge management	Single-cell Li-ion + TP4056-class charge IC	—	Power source and safe charging.
Fuel gauge	MAX17048-class coulomb counter	I2C	Accurate state-of-charge estimation (Feature 6), more reliable than raw voltage under load.
Integrity sensor	Conductive loop or reed switch in strap clasp	GPIO	Removal / tamper detection (Feature 11).
SOS input	Tactile momentary switch	GPIO (interrupt)	Manual emergency trigger (Feature 2).
Local feedback	Vibration motor + status LED	GPIO / PWM	Confirms SOS activation and device state to the participant without a screen.
Long-range radio (optional upgrade)	LoRa module, e.g. RFM95 / SX1276-class	SPI	Extends wearable-to-gateway range for spread-out trail deployments beyond BLE's effective reach.

For the MVP, ESP32's built-in BLE is recommended as the wearable-to-gateway link, with a dedicated LoRa module flagged as the deployment-specific upgrade path (Section 11) for use cases where participants may be spread across distances greater than BLE's typical open-air range.

3.6 System Implementation

1. Pre-event registration — devices are provisioned, paired to participant records, assigned to groups, and geofence zones are drawn for the event area.
2. Event-day check-in — an initial roster confirmation establishes that every registered device is online before the activity begins.
3. Continuous monitoring — throughout the activity, the system runs silently in the background, generating no dashboard noise for participants in the SAFE state.
4. Exception handling — as needed, following the alert flows described in Sections 17–18.
5. Scheduled checkpoints — attendance is verified at defined moments (Feature 7) without interrupting the activity for a full manual roll call.
6. Event close — a final headcount, incident log export, and device recovery.
7. Post-event — data is retained for a defined review window and then deleted, per the privacy policy in Section 23.

This is the default path, and by design it is the path taken almost all of the time:

1. Wearable samples its sensors on the adaptive schedule described in Sections 7 and 10.
2. On-device processing classifies activity state and checks for any locally-detectable exception.
3. Telemetry is transmitted on its normal interval to the gateway.
4. The gateway relays the packet to the backend.
5. The alert engine evaluates the update and confirms the participant remains SAFE.
6. The dashboard's LatestState reflects the update; no alert is generated and no organizer action is required.

That the dashboard usually shows nothing happening is the intended, correct behavior of an exception-based system — it is a sign the system is working, not a sign it has nothing to do.

Using an SOS/critical fall as the representative case:

1. Trigger — either a manual SOS press or an automatically detected possible fall.
2. The device buffers the event locally and assembles a high-priority packet, bypassing normal duty-cycle scheduling.
3. Prioritized transmission with retry-until-acknowledged, as described in Feature 2.
4. The backend flags the event CRITICAL immediately, bypassing normal batching.
5. The dashboard surfaces the alert prominently and requires explicit organizer acknowledgment before it can be cleared.
6. The organizer initiates a response following the guidance in the relevant feature section.
7. The resolution is logged with a timestamp and notes, closing the loop for later review and for the incident record described in Section 23.

Prototype MVP Scope

The specification requires a clear line between prototype capability and the fuller proposed system. That line is drawn feature by feature below.

Feature	MVP Prototype	Future Enhancement
Identification	QR-assisted registration; device ID lookup	NFC/RFID-assisted registration
SOS	Long-press button, retry-until-ack	Multi-modal trigger options
Location	Adaptive-interval GNSS, last-known fallback	On-device path history, indoor positioning aids
Movement / fall detection	Threshold-based heuristic (impact + orientation + stillness)	ML-based classifier trained on pilot field data
Heart rate	Periodic PPG sampling with signal-quality gating	Improved motion-artifact filtering
Battery	Fuel-gauge based percentage and threshold alerts	Predictive remaining-runtime estimation
Check-in	BLE proximity with QR fallback	NFC/RFID checkpoint option
Connection status	Heartbeat + timeout classification	Signal-strength-based early warning before full loss
Geofencing	Circular zones, backend-side evaluation	Polygon zones, on-device evaluation
Separation detection	Centroid-based, outlier-excluded	Per-activity auto-tuned thresholds
Wearable integrity	Conductive loop in strap clasp	Context-aware severity (scheduled vs. unscheduled removal)
Communication	BLE to gateway, Wi-Fi/cellular backhaul	LoRa option for spread-out deployments

Every row in the “Future Enhancement” column is explicitly out of scope for the initial prototype and is not assumed or implied anywhere else in this document.

3.7 Testing and Evaluation

1. Bench / unit testing per subsystem — GNSS fix acquisition time, IMU threshold behavior, HR signal quality, battery discharge-curve calibration — tested independently before integration.
2. Firmware integration testing on a single assembled unit, verifying the subsystems behave correctly together.
3. Controlled small-scale trial — a handful of devices in a known, safe area, with staff deliberately simulating falls, removals, SOS presses, and separations to measure both detection rate and false-positive rate.
4. Communication range and reliability testing at the actual intended venue — range depends heavily on terrain and obstruction, so this cannot be validated in the lab alone.
5. Organizer usability testing of the dashboard with non-technical staff, since the system's value depends entirely on organizers being able to act on it quickly under pressure.
6. A short supervised pilot at an actual small-scale camp or trip, run with standard manual safety procedures still active in parallel — the system is not relied upon exclusively until it has demonstrated itself in that pilot.

Fall detection, separation thresholds, HR reliability under motion, and integrity-loop durability — the same items flagged in Section 20 — are the features most in need of this real-world validation before their default thresholds should be trusted beyond the pilot stage.

3.8 Data Collection and Analysis

During the controlled small-scale trial and supervised pilot described in Section 3.7, detection rate and false-positive rate are logged per simulated event (fall, removal, SOS, separation) and per participant-hour of normal activity, so default thresholds can be adjusted with evidence before being trusted beyond the pilot stage. No participant data collected during testing is retained beyond the review window defined in Section 3.9.

3.9 Ethical Considerations

All wearable-to-server transmissions are encrypted (AES-128 as a floor, with TLS on the gateway-to-backend hop as a second layer). The data model is deliberately minimal: the system stores current state and a record of exceptions (Section 13), not a continuous raw location log, which limits both the privacy exposure and the storage burden. Data is scoped to the event and auto-deleted after a defined retention window rather than retained indefinitely. Dashboard access is role-based and limited to registered organizer accounts — participants and the general public have no access to the monitoring data.

As a matter of good practice, families and guardians of minor participants should be informed in advance that a monitoring device will be worn during the event, consistent with the transparency this kind of system depends on for trust. Specific legal or regulatory obligations around children's data — which vary by jurisdiction and by institution — are outside the scope of this technical proposal and should be reviewed with the organizing institution's own counsel before deployment rather than assumed from this document.

CHAPTER FOUR

Expected Results and Discussion

4.1 Expected System Output

At the end of the staged prototyping path (Chapter Three, Section 3.2), the expected output is a working MVP: a small fleet of wrist-worn wearables reporting to a gateway and backend, an organizer dashboard reflecting live SAFE/WARNING/CRITICAL/UNKNOWN status per participant, and a logged alert history covering the eleven core features defined in Section 3.3.1. These are expected outputs of a proposal at this stage, not results already obtained.

4.2 Expected Performance

Latency, detection-rate, and false-positive figures cannot be meaningfully stated before the bench, integration, and controlled-trial testing described in Section 3.7 has been carried out; the proposal deliberately avoids presenting invented numbers as though they were measured. Section 4.6 instead states the targets the pilot is designed to validate against.

4.3 Expected Benefits

The most direct impact is response time: an exception that previously might not surface until the next scheduled headcount can instead reach an organizer within seconds of being detected. The second, less visible impact is on organizer workload — the gap between overseeing a group of fifteen and a group of a hundred and fifty is, under manual methods, largely a matter of how much continuous attention a person can sustain; an exception-based system compresses that gap by removing the need to manually verify participants who require no attention at all. A third impact is a better incident record: timestamped alerts and logged responses produce a more complete account of what happened and when than memory and verbal reports alone.

The underlying pattern — a few supervisors responsible for many people across a physical area — is not unique to outdoor camps, and the same architecture could plausibly extend to adjacent settings in the future. This proposal, however, is deliberately scoped to organized outdoor groups within a defined event area, and makes no claim beyond that scope.

4.4 Expected Innovation

SAFE LINK's contribution is architectural rather than a claim of new sensing hardware: converting continuous multi-sensor telemetry into a small, fail-safe set of organizer-facing decisions, with connectivity loss treated as an explicit UNKNOWN state rather than assumed safety. This distinguishes it from both raw telemetry dashboards (which shift interpretation work onto the organizer) and general-purpose consumer trackers (which are not built around a group-event accountability workflow).

4.5 Expected Challenges and Solutions

Risk	Impact	Mitigation
GNSS unavailable (canopy, indoors, canyons)	Location shows LAST KNOWN instead of CURRENT	Dashboard always labels freshness explicitly; never silently reuse a stale fix
Battery life vs. sampling rate	Higher-fidelity monitoring drains the battery faster	Adaptive sampling (Sections 7, 10) trades resolution for runtime automatically

Risk	Impact	Mitigation
False positive alerts (fall, separation)	Organizer attention consumed by non-events	Multi-signal fusion, tunable thresholds, “possible” framing requiring verification
Communication range / dead zones	Wearable drops to OFFLINE despite being fine	Local + gateway buffering, resync on reconnect, honest UNKNOWN state rather than assumed SAFE
Integrity-loop durability under repeated flexing	False removal alerts, or missed real removals over time	Mechanical design review during prototyping; flagged for field validation (Section 24)
HR signal quality during motion	Frequent UNAVAILABLE/ABNORMAL readings while active	Signal-quality gating; treated as supporting signal only, never authoritative
Sensor/battery calibration drift over time and temperature	Gradually inaccurate battery percentage or fall thresholds	Periodic recalibration procedure; conservative default thresholds
Gateway as a single point of failure	All wearables behind a downed gateway appear OFFLINE simultaneously	Overlapping gateway coverage for larger events; dashboard should distinguish a mass-offline event from isolated ones

The system's core defense against false alerts is structural, not just a matter of tuning: nothing is ever reported as a confirmed event, only as a “possible” one requiring verification, and multi-signal fusion (Section 7, Feature 4) is used specifically to keep single-sensor noise from reaching the dashboard as an alert at all. Thresholds are tunable per activity type rather than fixed globally, because a single global threshold is itself a common source of false positives across genuinely different contexts (a hike versus a rest period, for example).

Failure modes are treated as design cases, not edge cases dealt with after the fact. A device failure (dead battery, hardware fault) results in an OFFLINE / UNKNOWN status, never a silent default to SAFE — the same fail-safe principle that governs Feature 8 throughout. A gateway or network failure triggers local buffering and resync rather than data loss, as described in Section 11, with the dashboard reflecting a degraded-but-not-silent state during the outage. A backend failure is the one case not otherwise covered above: the dashboard should display a visible system-health banner (“data last updated Xs ago”) whenever the backend itself stops updating, so organizers know to fall back to manual procedures rather than trusting a frozen screen that looks identical to a quiet, uneventful moment.

4.6 Evaluation Criteria

These are illustrative pilot-validation targets that the testing plan in Section 3.7 is designed to confirm or revise — not figures already demonstrated.

Criterion	Illustrative Target	Validated By
SOS event — organizer-visible acknowledgment latency	Within a few seconds of transmission, network permitting	Controlled trial (Section 3.7, item 3)
Possible-fall detection — false-positive rate	Low enough that organizers do not begin tuning out alerts	Controlled trial + pilot (Section 3.7, items 3 and 6)
Connection-loss classification — time to UNKNOWN state	Two to three heartbeat intervals, per Section 3.4.3	Bench / integration testing (Section 3.7, items 1–2)
Checkpoint check-in — coverage at a scheduled assembly	High automatic coverage, named-participant fallback for the remainder	Pilot (Section 3.7, item 6)

Criterion	Illustrative Target	Validated By
Organizer dashboard — usability under time pressure	Non-technical staff can identify and act on a WARNING/CRITICAL entry without training beyond a brief walkthrough	Usability testing (Section 3.7, item 5)

CHAPTER FIVE

Work Plan and Project Management

5.1 Project Activities

- Requirements review and proposal finalization.
- Subsystem bench testing — GNSS, IMU, PPG, fuel gauge, integrity loop (independently, per Section 3.7).
- Firmware development — sensor scheduling, on-device classification, local buffering, radio link.
- Backend and dashboard development — ingestion endpoint, alert engine, data API, organizer dashboard.
- Firmware/backend integration testing on a single assembled unit.
- Controlled small-scale trial with simulated falls, removals, SOS presses, and separations.
- Communication range and reliability testing at the intended venue.
- Organizer usability testing of the dashboard.
- Supervised pilot at an actual small-scale camp or trip, manual procedures active in parallel.
- Documentation, incident-record review, and proposal-to-prototype retrospective.

5.2 Work Breakdown Structure (WBS)

WBS Item	Deliverable
1. Planning	Finalized requirements, proposal sign-off
2. Hardware	Wearable BOM, breadboard-validated subsystems, assembled prototype unit(s)
3. Firmware	Sensor scheduling, on-device classification, local event buffer, radio link
4. Backend	Ingestion endpoint, alert engine, database, data API
5. Dashboard	Summary bar, participant list, map, alert center, participant detail page
6. Integration	End-to-end wearable-to-dashboard data flow on a single unit
7. Testing	Controlled trial results, range testing results, usability testing notes
8. Pilot	Supervised field pilot report and threshold recommendations

5.3 Project Schedule

An illustrative six-week summer-camp schedule, sequenced so that no integration step begins before its constituent subsystems have been independently validated (Section 3.2):

Activity	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6
Planning & requirements	✓					
Subsystem bench testing	✓	✓				
Firmware development		✓	✓			

Activity	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6
Backend & dashboard development		✓	✓			
Integration testing			✓	✓		
Controlled trial & range testing				✓	✓	
Usability testing					✓	
Supervised pilot & documentation						✓

5.4 Milestones

- M1 — Proposal finalized and subsystem bench testing complete.
- M2 — Firmware and backend feature-complete against the MVP scope (Chapter Three, Table 7).
- M3 — Single-unit end-to-end integration demonstrated.
- M4 — Controlled-trial detection and false-positive rates recorded.
- M5 — Supervised pilot completed and threshold recommendations documented.

5.5 Team Responsibilities

Roles below are illustrative and should be assigned by the summer-camp team according to member interest and prior experience:

Role	Primary Responsibility
Hardware lead	Component sourcing, breadboard validation, enclosure and strap design
Firmware lead	Sensor scheduling, on-device classification, radio link, local event buffer
Backend / dashboard lead	Ingestion endpoint, alert engine, database, organizer dashboard
Testing & QA lead	Test-case design, controlled-trial coordination, results logging
Camp mentor / advisor	Technical review, milestone sign-off, pilot supervision

CHAPTER SIX

Budget and Resources

6.1 Hardware Cost

Per-unit component cost at prototype quantities is difficult to state precisely given fluctuating supplier pricing, and is illustrative only — it should be validated against current supplier quotes before any build-out quantity is committed. The figures below assume a single-wearable BOM at hobbyist quantities, dominated by the GNSS and communication modules, consistent with the feasibility discussion in Chapter Three, Section 3.2.

No.	Item	Qty	Illustrative Unit Cost (USD)	Illustrative Total
1	ESP32-class microcontroller board	1	3–6	3–6
2	GNSS receiver (NEO-6M/M8N-class)	1	6–12	6–12
3	IMU (MPU6050-class)	1	1–3	1–3
4	PPG heart-rate sensor (MAX30102-class)	1	3–6	3–6
5	Fuel gauge IC (MAX17048-class)	1	2–4	2–4
6	Li-ion cell + TP4056-class charge IC	1	3–5	3–5
7	LoRa module (optional upgrade)	0–1	5–9	0–9
8	SOS button, vibration motor, LED, strap-integrity loop	1 set	2–4	2–4
9	Enclosure and strap	1	2–5	2–5
	Illustrative total per wearable			≈ 25–55

6.2 Software Cost

- Firmware development — open-source toolchains (ESP-IDF / Arduino framework); no license cost.
- Backend — a single small managed cloud instance and database is sufficient at MVP scale (tens to a few hundred devices); many providers offer a free or low-cost tier adequate for a prototype and pilot.
- Dashboard — built on open-source web frameworks; no license cost anticipated for the MVP.

6.3 Human Resources

The prototype is designed to be delivered by the student summer-camp team described in Chapter Five, Section 5.5, with technical review and milestone sign-off from assigned camp mentors. No paid external labor is assumed for the MVP stage.

6.4 Other Expenses

Item	Notes
Custom PCB fabrication (post-breadboard stage)	Low-quantity prototype run, per Chapter Three, Section 3.2

Item	Notes
Venue / field-test logistics	Controlled trial and supervised pilot, per Section 3.7
Contingency	Illustrative 10–15% buffer against component price fluctuation

6.5 Resource Requirements

- Basic electronics lab / workbench access for breadboard and PCB assembly work.
- Development boards for each subsystem (GNSS, IMU, PPG, fuel gauge) ahead of integration.
- A small managed cloud instance and database for the backend.
- Reliable internet access at the development site; venue backhaul should be surveyed separately before the pilot, per Chapter Three, Section 3.4.3.
- Standard firmware and web development tooling (IDE, version control, basic CI).

CHAPTER SEVEN

Conclusion

7.1 Summary

SAFE LINK reframes participant safety monitoring from continuous manual vigilance to exception-based response, built from feasible, well-documented embedded components and a clearly scoped MVP that separates what a prototype can deliver now from what remains future work. Every feature in this document connects a specific hardware signal to a specific organizer decision, from the SOS button through to wearable-integrity detection, and every claim about accuracy, range, or reliability is stated with its actual limitations rather than presented as guaranteed.

The recommended next step is to move from this proposal into the staged prototyping path described in Section 20: subsystem-level validation first, followed by integration, followed by the controlled and pilot testing described in Section 24 — building the evidence needed to trust the system's thresholds before it is relied upon in an actual event.

7.2 Expected Contribution

This proposal contributes a clearly scoped, evidence-driven MVP architecture for organized-group outdoor safety monitoring: an eleven-feature exception-based system built entirely from feasible, well-documented embedded components, with an explicit line drawn between what the prototype delivers now and what remains future work (Chapter Three, Table 7). Its primary value to the summer-camp program is a working demonstration that a small student team can carry a safety-relevant embedded system from bench-tested subsystems through a supervised field pilot within a single camp cycle.

7.3 Future Work

- Solar-assisted charging for multi-day trips.
- ML-based fall and activity classification trained on pilot field data, to reduce false-positive rates below what fixed thresholds can achieve.
- NFC/RFID checkpoint option alongside BLE proximity check-in.
- Polygon geofencing with on-device evaluation.
- Satellite backhaul for fully remote deployments — flagged explicitly as requiring dedicated budget and hardware evaluation, not assumed feasible within the current scope.
- Integration hooks for existing camp-management or registration software.
- Multi-gateway redundancy for larger events.
- Institution-configurable data-retention policy settings.

References

Component references follow the IEEE citation style, appropriate for an embedded-systems / engineering proposal. Entries below are manufacturer datasheets for the illustrative component classes discussed in Chapter Three; exact part numbers and revisions should be confirmed against the supplier's current documentation before procurement.

[1] Espressif Systems, “ESP32 Series Datasheet,” Espressif Systems, Shanghai, China.

[2] u-blox AG, “NEO-M8 Series u-blox M8 Concurrent GNSS Modules — Data Sheet,” u-blox AG, Thalwil, Switzerland.

[3] TDK InvenSense, “MPU-6000 and MPU-6050 Product Specification,” TDK InvenSense, San Jose, CA, USA.

[4] Analog Devices (Maxim Integrated), “MAX30102 High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health — Data Sheet,” Analog Devices, Inc., Wilmington, MA, USA.

[5] Analog Devices (Maxim Integrated), “MAX17048/MAX17049 3µA 1-Cell/2-Cell Fuel Gauges with ModelGauge — Data Sheet,” Analog Devices, Inc., Wilmington, MA, USA.

[6] Semtech Corporation, “SX1276/77/78/79 — 137 MHz to 1020 MHz Low Power Long Range Transceiver Datasheet,” Semtech Corporation, Camarillo, CA, USA.

[7] Bluetooth SIG, “Bluetooth Core Specification,” Bluetooth Special Interest Group, Kirkland, WA, USA.

[8] Nordic Semiconductor / IEEE, general references on low-power wireless sensor network design, consulted for the communication-architecture discussion in Chapter Three, Section 3.4.3.

Note: this proposal deliberately does not cite specific accuracy, range, or false-positive figures as though independently published, since those values are the object of the testing plan in Chapter Three, Section 3.7, rather than established facts drawn from prior work.

Appendix A — Sample Organizer Dashboard Alert Gallery

A consolidated, side-by-side view of the alert card types referenced throughout Chapter Three, Section 3.3.1, shown together here for quick visual reference.

ORGANIZER DASHBOARD VIEW — CRITICAL — SOS ALERT

```
⚠ CRITICAL SOS ALERT
Participant: Participant 023
Location: [Current or Last Known Coordinates]
Status: ASSISTANCE REQUIRED
```

ORGANIZER DASHBOARD VIEW — CRITICAL — GEOFENCE EXIT

```
GEOFENCE ALERT
Participant: Participant 023
Event: Exited Safe Zone
Time: 10:42:16
```

ORGANIZER DASHBOARD VIEW — WARNING — PROLONGED INACTIVITY

```
WARNING: PROLONGED INACTIVITY
Duration: 18 minutes
Last Known Location: [Coordinates]
```

ORGANIZER DASHBOARD VIEW — WARNING — CONNECTION LOST

```
STATUS: UNKNOWN – CONNECTION LOST
Last Contact: 10 minutes ago
Last Known Location: [Coordinates]
```

ORGANIZER DASHBOARD VIEW — NORMAL — PARTICIPANT IDENTIFICATION

```
Participant: Participant 023
Device ID: WG-023
Status: NORMAL
```

Appendix B — Component-to-Datasheet Cross-Reference

Maps each hardware component class discussed in Chapter Three to its corresponding reference entry.

Component Class	Reference Datasheet	Reference No.
Microcontroller (ESP32-class)	Espressif ESP32 Series Datasheet	[1]
GNSS receiver (NEO-6M/M8N-class)	u-blox NEO-M8 Series Data Sheet	[2]
IMU (MPU6050-class)	TDK InvenSense MPU-6050 Product Specification	[3]
PPG heart-rate sensor (MAX30102-class)	Analog Devices MAX30102 Data Sheet	[4]
Fuel gauge (MAX17048-class)	Analog Devices MAX17048/49 Data Sheet	[5]
LoRa radio module (SX1276-class)	Semtech SX1276/77/78/79 Datasheet	[6]
BLE radio (integrated)	Bluetooth Core Specification	[7]